

CLASS-12

SENIOR SECONDARY EXAMINATION -2015

BIOLOGY

TIME: 3 ¼ HOURS

MM: 56

GENERAL INSTRUCTION TO THE EXAMINEES.

1. Candidates must write first his/ her roll number on the question paper compulsorily.
2. All questions are Compulsory.
3. Write the answer to each question in the given answer book only .
4. For questions having more than one part the answers to those parts are to be written together in continuity.
5. If there is any error difference contradiction in Hindi and English versions of the question paper the questions of Hindi version should be treated valid.
6. For questions having more than one part the answers to the other part should be written together in continuity.

1. Section	q.no	marks
A	1-13	1
B	14-24	2
C	25-27	3
D	28-30	4

7. There are internal services in question 24, 27, 28, 29, 30.

SECTION-A

1. Why do pollen grains remain preserved for long time ? [1]
Ans: Pollen Grain remain preserved for long time for the cross pollination. It is a device to bring about genetic recombination and variation.
2. What are stem cells ? [1]

Ans: Stem cells are primal cells found in all multi-cellular organisms. They retain the ability to renew themselves through mitotic cell division and can differentiate into a diverse range of specialized cell types.

3. **what is Sickle Cell anaemia ? [1]**

Ans: sickle cell disease (SCD), is a genetic disease of the red blood cells (RBCs). RBC formed during the maturation are sickle shaped due to low oxygen condition and they are partially functional.

4. **What is the exception to Mendel's law of independent assortment ? [1]**

Ans: Linkage is exception to Mendel's Law of independent assortment .

5. **Define genetic drift. [1]**

Ans: Random changes in the allele frequencies of a population occurring only by chance constitute genetic drift.

6. **Write the name of leaf and stick resistant variety of wheat. [1]**

Ans: Himgiri is leaf and stick resistant **variety of wheat**.

7. **Penicillin is obtained from which fungus? [1]**

Ans: Penicillin is obtained from fungus *Penicillium notatum*.

8. **Give the name of that bacterial product which is used as a clot buster. [1]**

Ans: Streptokinase is a chemical used as a blockbuster to dissolve the blood clot in the blood vessels of patients suffering from myocardial infarction .It is obtained from *Streptococcus* bacteria.

9. **Write the names of two endonuclease enzyme. [1]**

Ans: *ECORI*, *HIND ii*

10. **Write the names of two antibiotic resistance genes found in plasmid PVR 322. [1]**

Ans: Ampicillin resistant gene and Tetracycline resistance gene.

11. **Where is the International Centre for wheat and Maize situated. [1]**

Ans: The International Centre for wheat and Maize situated in Mexico.

12. **Give the names of four greenhouse gases present in the atmosphere. [1]**

Ans: Carbon dioxide, Methane, chlorofluorocarbon, nitrous oxide are four greenhouse gases present in the atmosphere.

13. **What is accelerated Eutrophication? [1]**

Ans: The increased productivity of lake and pond etc. due to nutrient enrichment is called accelerated Eutrophication.

SECTION – B

14. Why can be a offspring produced by asexual reproduction called clone. Give reason.

Ans: The term clone is referred to describe morphologically and genetically similar individuals derived from the same cell. In a sexual reproduction offspring's are not genetically identical to one another but exact copies of their parents. Offspring produced by asexual reproduction are exact copies of their parents. Such offspring's are genetically also identical.

15. If acrosome of sperm is damaged then which process will be affected and how? [2]

Ans: Acrosome is the head regions of sperm that contains various penetrating enzymes for example that helps sperm to enter into the cytoplasm of egg. During fertilization if the acrosome is damaged the sperm will never be able to penetrate ovum.

16. What suggestions will you give to in fertilized couples to produce children? [2]

Ans: There are various methods that help couples to produce child.

- Test tube baby program.
- Gamete Intra fallopian transfer (GIFT).
- Intracytoplasmic sperm injection (ICSI)
- Artificial insemination

17. Give four adaptation found in animals of arid region. [2]

Ans: Adaptations in arid animals are following-

- Animals have High deposition of fat like in hump of camel.
- Animals living in desert area like kangaroo and camel release high concentrated urine and clear the influence of ADH secretion.
- Desert lizard and snake can hide under the shades of rocks during daytime.
- Few species are capable of growing into soil to hide and escape from above ground heat.

18. Wasp prefer inflorescence of fig tree for laying eggs. Explain giving reasons. [2]

Ans: Wasp preferred inflorescence of fig tree for laying its eggs because during the hatching of the larva. It feeds upon some of the developing seeds of the big fruit.

19. As a student of Biology how will you explain to farmers that bee keeping is simple economically beneficial for them. [2]

Ans: Bee keeping can be simple and economically beneficial for farmer in many aspect

- The products of honey bees are Honey with which has high food value and medicinal importance.
- Bee wax are used in cosmetics, Paints and polishes .
- Honey bees are of chief cross pollination agents hence they increase the pollinating efficiency which can further that improve the yield of crop.

20. What will be the effect on treatment of sewage if microbes are removed from it ? Clarify. . [2]

Ans: During the secondary treatment of sewage there is increase in the air supply to accomplish the digestion of organic matter by bacteria present in it which oxidize the organic matter into CO_2 and H_2O so if microbes from the sewage is removed the organic matter in it not be digested and B. O. D. value hence will not be decreased.

21. Write four goals of Human Genome Project. . [2]

Ans: Important goal of HGP are-

- Identifying the 20000 to 25000 approx change in human DNA .
- Determining the frequency of the 3 billion chemical base pairs of the human DNA.
- Store this information in database
- Improve tools for data analysis.
- Transfer related Technologies to other sectors like Industries .
- Address the ethical legal and social issues(E LSI) that can arise from the project.

22. Explain the role of transgenic animals in production of biological products. . [2]

Ans: Due to increase in the population the demand for various products like insulin, tissue ,plasminogen activator ,lactoferrin blood clotting factor etc. has also been increased. Genetically engineered insulin that is humulin is produced by generics animals. for example –

1.Production of insulin from transgenic E coli.

2.Production of human protein enriched milk by transgenic cow, Rosie .The milk contains human Alpha lactalbumin which is much more balance product for women baby than natural cow milk.

23. What suggestions will you give to save an imbalanced ecosystem. . [2]

Ans: Ecosystem serves various environmental aesthetic and economic values which can only be obtained if we imply necessary steps to save an imbalance ecosystem.

24. What is adaptive radiation.explain it with an example.

1+1=2

Ans: Adaptive radiation is an evolutionary process in which an ancestral stock gives rise to new species adapted to new habitats and new way of life. Examples of adaptive radiation-

- Darwin's finches-These were small black Birds which Darwin observed in Galapagos Islands.
- These were many varieties in the same Island.
- He isn't that after originating from a common ancestor seed eating stock and finches must have radiated to different geographical areas and undergone adaptive changes specially in the type of beaks.
- living in isolation for long the new kinds of finches in words that could function and survive in the new habitat

OR

Explain homology and analogy in animals by giving examples. [2]

Ans:

- Organs different in function but of similar embryonic origin and development have similar relationships with adjacent organ is known as homology and the organs are known as homologous organ for instant arms of man, leg of horse, wing of a bird or a bat, flipper of a seal and forelimb of a cheetah are apparently different from one another in structure and function yet all of them are build on one basic plan, the pentadactyl plan.
- Analogy and analogous organs -Analogous organs are those which perform the same function has superficial resemblance to one another but are of different origin and this phenomena is called analogy for example Wings of insects ,Birds and bats performs the same function of flying and possess superficial resemblance but their basic structure is entirely different.

25. What is endosperm? Explain various types of endosperm .

1+2=3

Ans: The fusion of the central cell and the second male gamete that is triple fusion gives rise to the structure called endosperm which provides nutrition to the growing embryo. The various types of endosperm are –

1. Nuclear Endosperm- The primary endosperm nucleus divides repeatedly without wall formation to produce a large number of free nuclei.
2. Cellular Endosperm- In the constitutive division of the primary endosperm every karyokinesis followed by cytokinesis which gives rise to cellular endosperm.
3. Helobial Endosperm – In this division of primary endosperm

26. What is Co-dominance? Explain with example.

1+2=3

Ans: When two alleles of a trait equally express with neither recessive nor dominant it creates co-dominance. ABO blood group system is a good example of co-dominance. The gene 'I' for the blood group contains three alleles I^A , I^B and I which represent the

phenomena of multiple allelism. as it has more than two alleles governing the same character. The alleles I^A and I^B are co-dominant .They express themselves even in the presence of the other .Thus the presence of both the alleles I^A and I^B gives the blood group of AB.

Genetic Basis of Blood Groups in Human Populatio			
Allele from Parent 1	Allele from Parent 2	Genotype of Offsprings	Blood types of Offsprings
I^A	I^A	$I^A I^A$	A
I^A	I^B	$I^A I^B$	AB
I^A	i	$I^A i$	A
I^B	I^A	$I^A I^B$	AB
I^B	i	$I^B i$	B
i	i	ii	O

27. What is Recombinant DNA.Explain the process of separation and isolation of DNA fragments by gel electrophoresis technique.

IT is the DNA formed by combining DNA from two different organisms .

Separation and isolation of DNA fragment-

- The cutting of DNA by restriction endonuclease results in the fragment of DNA.Fragments can be separated by the technique known as gel electrophoresis.
- Since DNA fragments are negatively charged molecules they can be separated by forcing them to move towards the anode under and electric field through a medium .
- Now a days the most commonly used matrix is agarose which is a natural polymer extracted from sea weeds.
- The DNA fragments separate according to their size through sieving effect provided by the agarose gel Hence smaller the fragments size the farther it moves.
- The separated DNA fragments can be visualized only after changing the DNA with a compound known as ethium Bromide followed by exposure to UV radiation. Bright Orange coloured bands of DNA can be seen in aethium Bromide gel exposed to UV light.
- The separated band of DNA are cut off from the agarose gel and extracted from gel piece. This state is known as elution.
- The DNA fragments purified in this way are used in constructing Recombinant DNA by joining them with cloning vectors.

OR

Ans: Genetic engineering is the technique used to alter the chemistry of genetic material which are DNA and RNA. This is done to introduce the genetic material into host organism and thus change the phenotype of host organism.

Amplification refers to the process of making multiple copies of the DNA segment in Vitro. It employs Polymerase Chain Reaction. The process was designed by K. Mullis. This technique involves main 3 steps-

Denaturation

Primer annealing

extension of primers

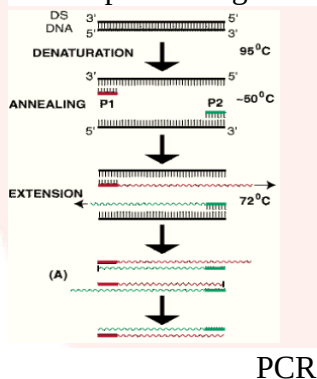
The double stranded DNA is denatured by using high temperature.

Two sets of primers are used. Primers are the chemically synthesized short segment of DNA oligonucleotides that are complementary to the segment DNA.

DNA polymerase is the enzyme used to make copies of DNA making use of a genomic template DNA and the primer.

If the process of replication of DNA is repeated many times the segment of DNA can be amplified to the approximately billion times i.e. 1 billion copies are made. Such repeated amplification is achieved by the use of thermostable DNA polymerase isolated from bacterium *Thermusaquaticus*. It remains active during high temperature induced denaturation of double stranded DNA.

The amplified fragment if desired can now be used to ligase with a vector for the cloning.



PCR

SECTION- D

28. What is nucleosome draw diagram of double stranded polynucleotide chain of DNA and explain its structure.

Ans: It is made of two polynucleotide chains where the back bone is constituted by sugar phosphate and the bases project inside.

The two chains have antiparallel polarity it means if one chain has clarity $5' \rightarrow 3'$ The other has $3' \rightarrow 5'$.

The bases in two strands are paired through hydrogen bonds forming base pairs. Adenine forms hydrogen bonds with a thymine from opposite stand and vice-versa. Similarly cytosine is bonded guanine with three hydrogen bonds. As a result always Purine comes opposite to a Pyrimidine. This generates approximately uniform distance between the two strands of the helix.

The two chains are coiled in a right handed fashion. the pitch of The Helix is 3.4 nanometer (which is one billionth of a meter that is 10^{-9} m) and there is roughly 10 BP in each turn. Consequently the distance between a BP in helix is approximately equal to 0.34nm.

The plane of one base pair stacks over the other in a double helix thus in addition to H bonds confers stability of the helical structure.

OR

Write properties of genetic materials .Explain with diagram the experiment of Alfred Hershey and Martha chase that DNA is a genetic material . .

1+2+1=4

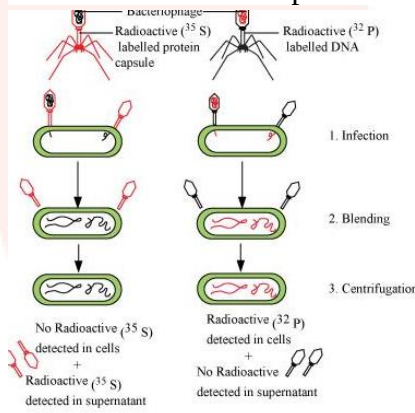
Ans: Properties of genetic materials are as follows-

It should be able to generate its replica (replication).

It should chemically and structurally stable.

It should provide the scope for slow changes (Mutation) that I required for evolution.

It should be able to express itself in the form of Mendelian characters.

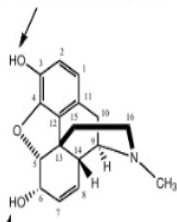


ALFRED HERSHEY AND CHASED EXPERIMENT.

29. From which plant is cocoa alkaloid for cocaine is obtained? What is its effect on human body? Write the chemical structure of morphine.

Ans: Coca alkaloids are obtained from erythroxylian Cocoa.

Cocaine interferes with the transport of neurotransmitter, dopamine. It has a potential stimulating action on the central nervous system and produce sense of euphoria and increase energy. Excessive dosage cause hallucination.



MORPHINE

OR

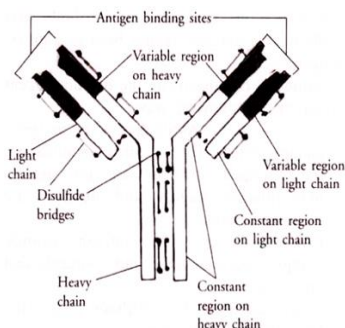
What is immunity ? Differentiate between active immunity and passive immunity

. Draw the diagram of structure of antibody molecule .

1+2+1=4

Ans: The overall ability of host to fight disease causing organism conferred by immune system is called immunity.

Active Immunity	Passive immunity
When the antibodies are developed by our self in the response to antigen, it is called active immunity.	When antibodies develop in other vertebrates In response to deliberate injection of antigen, are injected in our body, it constitutes passive immunity.
It takes time to develop immunity process which states It stays for longer period for example immunity developed during natural exposure to pathogen.	It is used when the immune response has to be faster.
It takes time to develop immunity process which states It stays for longer period. For example immunity developed during natural exposure to pathogen	It stays for short period . For example- Immunity given to infant by antibodies in colostrum.



STRUCTURE OF ANTIBODY

30. Define food chain how many types of food chain are there give differences between Food Chain and Food Web Royal area diagram of food chain.

Transfer of food energy from the producers through a series of organism(herbivores to carnivores to decomposers) with repeated eating and being eaten is known as food chain .In nature we generally distinguish the two general types of food chain.Grazing Food Chain and Detritus Food Chain.

The sequence of living organism in an ecosystem in which one organism consumes another organism to transfer food energy is called food chain there at the interconnected food chains operating in an ecosystem which establish a network of relationships between various species is called food web. Food web is a collection of food chains.

Linear food chain

Grass → Grasshopper → Rat → Snake → Hawk

OR

What is ecosystem? Explain the structure of ecosystem and draw a diagram of tropical level of ecosystem.

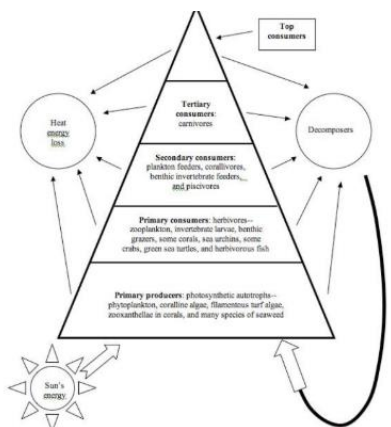
1+2+1=4

Ecosystem is a basic functional unit of organism and their environment, interacting with each other and with their own components. An Ecosystem is a self-sufficient units it may be very small for example of a freshwater pond or very large for example the Sahara desert.

The two important structural features of an ecosystem are-

Species composition- Species composition refers to all plants, animals and microbial species present in the given ecosystem .

Stratification - Stratification refers to the vertical distribution of different species of to find different levels in the ecosystem. for example trees occupy the top vertical layer of the forest. Shrub occupy the second layer and herbs occupied the bottom layer.



TROPICAL LEVEL OF ECOSYSTEM
